6.3 Ecological adaptations

The modifications in the structure of organisms to survive successfully in an environment are called **adaptations** of organisms. Adaptations help the organisms to exist under the prevailing ecological habitat. Based on the habitats and the corresponding adaptations of plants, they are classified as hydrophytes, xerophytes, mesophytes, epiphytes and halophytes.

Hydrophytes

The plants which are living in water or wet places are called hydrophytes. According to their relation to water and air, they are sub-divided into following categories: i) Free floating hydrophytes, ii) Rooted- floating hydrophytes, iii) Submerged floating hydrophytes, iv) Rooted -submerged hydrophytes, v) Amphibious hydrophytes.

i. Free floating hydrophytes: These plants float freely on the surface of water. They remain in contact with water and air, but not with soil. Examples: *Eichhornia, Pistia and Wolffia* (smallest flowering plant).

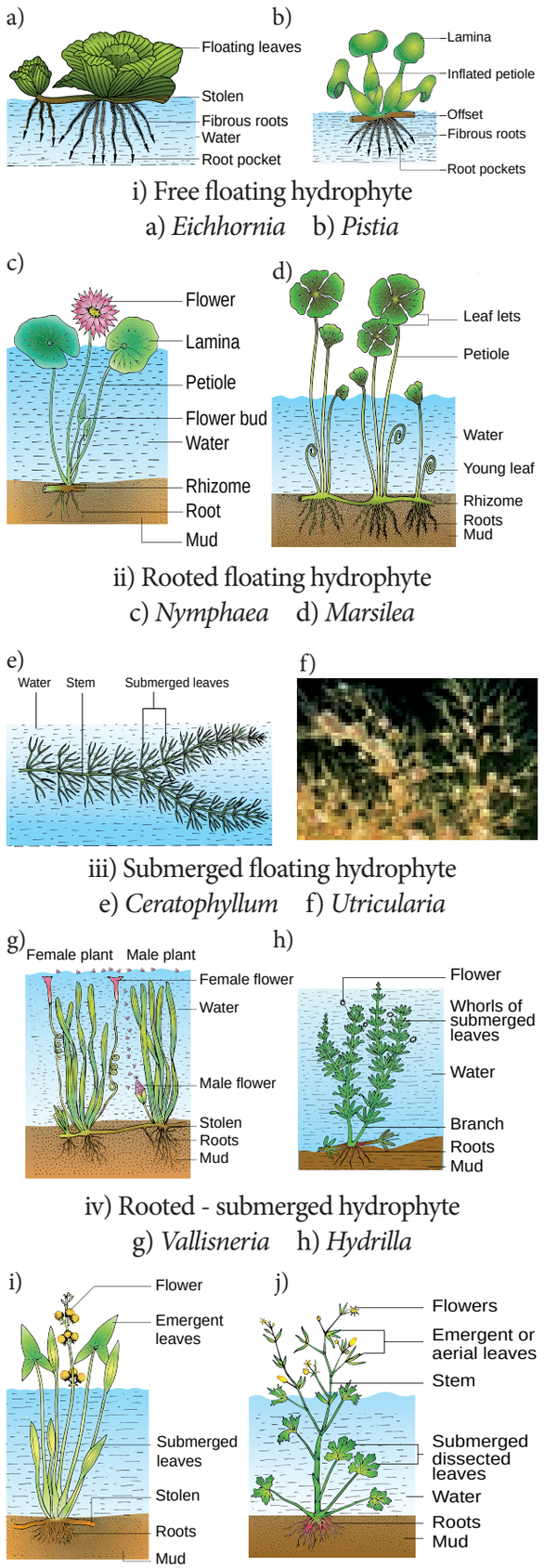
ii. Rooted floating hydrophytes: In these plants, the roots are fixed in mud, but their leaves and flowers are floating on the surface of water. These plants are in contact with soil, water and air. Examples: *Nelumbo, Nymphaea, Potomogeton* and *Marsilea.*

Lotus seeds show highest longevity in plant kingdom.

iii. Submerged floating hydrophytes: These plants are completely submerged in water and not in contact with soil and air. Examples: *Ceratophyllum* and *Utricularia.*

iv. Rooted- submerged hydrophytes: These plants are completely submerged in water and rooted in soil and not in contact with air. Examples: *Hydrilla*, *Vallisneria* and *Isoetes*.

v. Amphibious hydrophytes (Rooted emergent hydrophytes)**:** These plants are adapted to both



i) Free floating hydrophyte
a) *Eichhornia* b) *Pistia*

ii) Rooted floating hydrophyte
c) *Nymphaea* d) *Marsilea*

iii) Submerged floating hydrophyte
e) *Ceratophyllum* f) *Utricularia*

iv) Rooted - submerged hydrophyte
g) *Vallisneria* h) *Hydrilla*

v) Rooted emergent hydrophyte-Heterophylly
i) *Sagittaria* j) *Ranunculus*

Figure 6.18: Hydrophytes

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